Pack Smart, Schedule Fair

Unifying Heterogeneity and Fragmentation Awareness in GPU Scheduling

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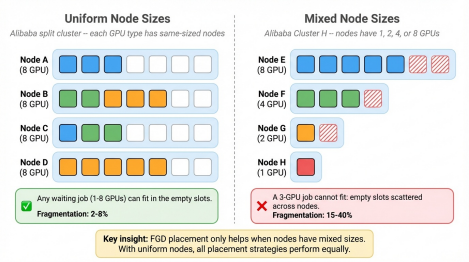
GPU Clusters Waste Capacity Through Fragmentation

GPU sharing splits GPUs across multiple tasks to improve utilization.

But it creates **fragments**: leftover GPU capacity too small for any waiting job.

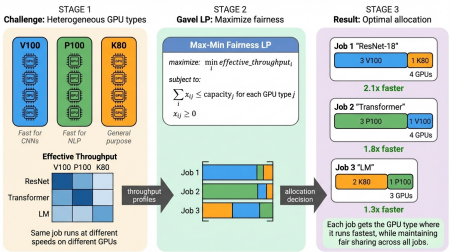
21–42% of free GPU capacity is unusable



- Average GPU utilization: 25–50%
- Fragments cannot consolidate across GPU boundaries
- Equivalent to hundreds of idle GPUs at cluster scale

Two Systems, One Gap

Gavel *(OSDI 2020)*



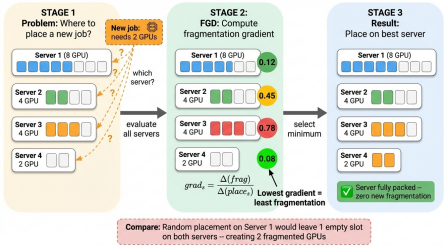
Solves LP to allocate jobs across GPU types.
Reduces JCT by 20–70%.

Gap: Ignores *where* jobs land.

+

?

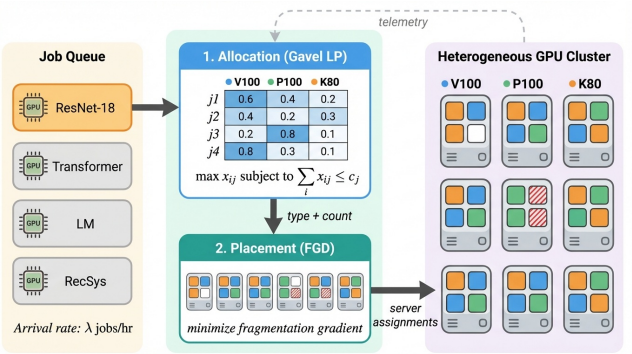
FGD *(ATC 2023)*



Places jobs to minimize fragmentation gradient.
Reduces wasted GPUs by 49%.

Gap: No flexible policy framework.

Our Approach: Two-Stage Architecture



Stage 1 – Allocation (Gavel):
LP computes time fractions per GPU type.
"Which type, how much?"

Stage 2 – Placement (FGD):
Selects node minimizing fragmentation gradient.
"Which specific node?"

Evaluation: Two Clusters, One Variable

Property	Alibaba Split	Cluster H
Total GPUs	6,212	5,592
GPU types	12 (heterogeneous)	1 (homogeneous)
GPUs/node	**8 (uniform)**	**1, 2, 4, 8 (mixed)**

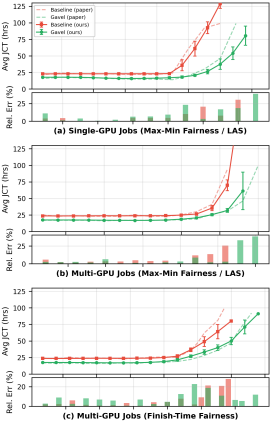
900 total experiments

- 15 arrival rates
- 3 random seeds
- 4 placement strategies: Random, Strided, BestFit, FGD
- 2 allocation policies: Max-Min Fairness, FIFO

Controlled experiment:
Same traces, same simulator, same placement strategies.
Only **node topology** differs.

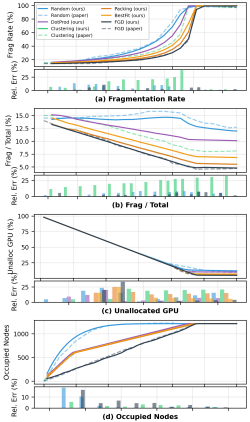
Replication Validates Both Baselines

Gavel (504 experiments, Philly trace)



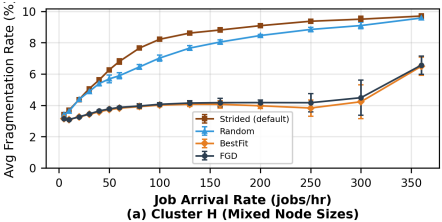
Solid: ours. Dashed: paper reference.
Close match at sub-saturation loads.

FGD (Alibaba Cluster H, 6 policies)

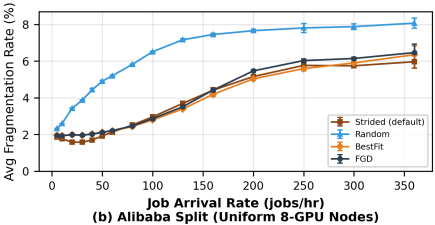


At 80% demand: Random fragments 85% of free GPUs vs. FGD's 42%.

FGD Reduces Fragmentation 30–40% on Mixed Nodes



(a) Cluster H (Mixed Node Sizes)



(b) Alibaba Split (Uniform 8-GPU Nodes)

Cluster H (mixed 1/2/4/8-GPU nodes)

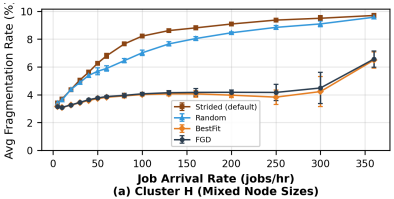
FGD and BestFit reduce fragmentation by **40–50%** vs. Strided across all arrival rates.

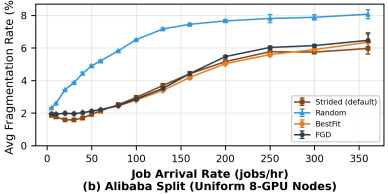
Consistent ordering:
FGD $\approx$ BestFit $<$ Random $<$ Strided

Holds under **both** Max-Min Fairness and FIFO.

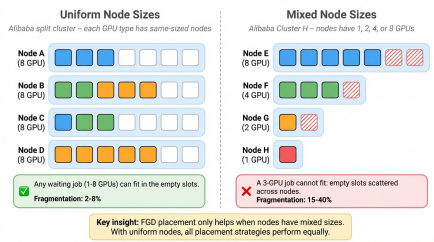
Placement benefit is independent of allocation policy.

But Zero Benefit on Uniform Nodes



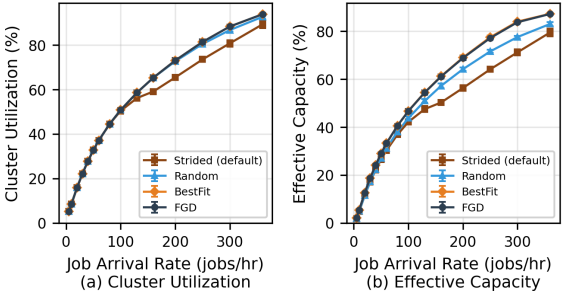


(a) Mixed nodes: curves spread
(b) Uniform nodes: curves converge



Fragmentation depends on *node topology diversity*, not GPU type heterogeneity.

Practical Impact: 280 GPUs Recovered



On Cluster H (mixed nodes):

280 recoverable GPUs

17% more completed jobs

- Strided caps at 89% utilization
- FGD reaches 94%
- Effective capacity gap: **13 points** at high load

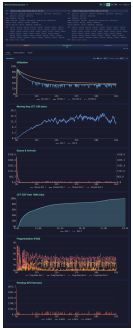
Per-job JCT identical across placements. Placement shapes capacity; allocation shapes throughput.

Limitations & Visualization Tool

Honest caveats:

- FGD is **sensitive to workload ordering** – underperforms in 5/7 arrival orders
- All results are **trace-driven simulation** – no feedback effects
- **Migration costs** not modeled

Interactive Visualization Tool:



843 experiments. Side-by-side comparison, time-series charts, aggregate results with paper overlays. Open source.

Fragmentation is driven by *node topology diversity*, not GPU type heterogeneity.

1. **Uniform clusters** don't need fragmentation-aware placement
2. **Mixed-hardware clusters** benefit substantially (30–40% less fragmentation, 17% more throughput)
3. **Allocation and placement are orthogonal** – compose them independently

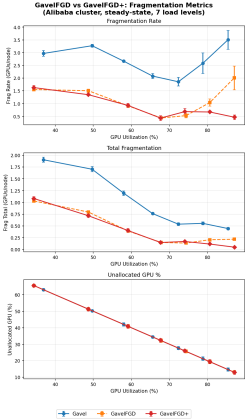
Simulator: `github.com/Turquoise-T/cs244c-GPU_fragment`
Viz tool: `github.com/varunr89/gpu-scheduling-viz`

Backup: Arrival Order Sensitivity

Task arrival order	FGD	Clustering	Random
trace	480.0	205.5	522.0
ascending	2167.0	835.0	1029.0
descending	110.2	129.0	107.6
phased 01234	1890.0	808.0	993.0
phased 12034	1318.0	332.0	925.0
phased 24031	949.0	162.0	586.0
phased 31042	124.0	267.9	421.7

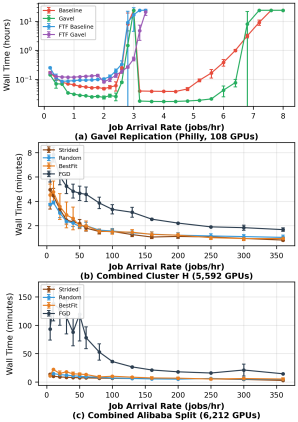
Total pending task GPUs at 96% arrived workload (Alibaba, 90% cluster scale). FGD underperforms Clustering in 5 of 7 orderings. Wins only when large jobs arrive first.

Backup: GavelFGD+ (POA Coordination)



At 85% utilization, GavelFGD+ reduces fragmentation by 76.7% relative to GavelFGD. Improvement comes from more efficient packing, not additional resources.

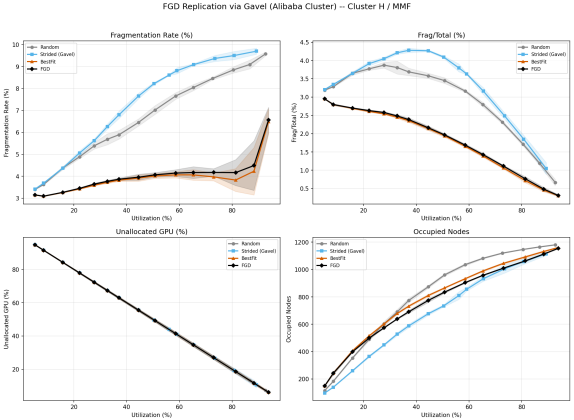
Backup: Wall Time Analysis



900+ experiments on Stanford FarmShare (Intel Xeon, no GPU required).

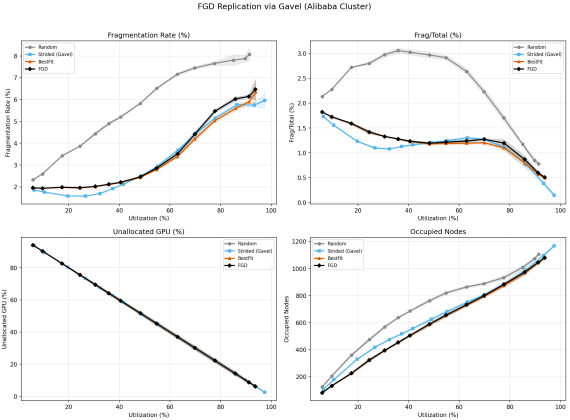
Philly: exponential near saturation. Cluster H: flat 1–6 min. Alibaba: decreasing with load.

Backup: Per-Utilization Cluster H



FGD (black) reduces fragmentation by 30–40% vs. Random (blue) at moderate utilization.
BestFit (orange) is intermediate. Strided (brown) tracks Random.

Backup: Per-Utilization Alibaba Split (Uniform)



All placement strategies converge. Node topology uniformity eliminates the fragmentation problem.